ILT Research System - Detailed Design Specification (DDD v1.3 CODING-READY)

Autonomous AI Coding Blueprint (Gemini & Claude Implementation Guide): Component Architecture, Wire Protocols, WebRTC Signaling Contracts, Seed DDL Scripts & Verification Test Harness

Document Parameter	Specification Value	Document Parameter	Specification Value
Document Identifier:	ILT-DDD-V1.3-CODING-READY-EN	Publication Date:	October 2026 (AI-Agent Implementation Edition)
Functional Baseline:	FRS v3.3 & Architecture Spec v5.3	Coding Agents:	Gemini & Claude (Parallel Autonomous Implementation)
Test Environment:	Local Docker Compose Stack + P2P LAN (Google Drive Not Usable)	Target Hardware:	Wacom MovinkPad 11 (Certified Continuous Charging)
Readiness Status:	100% Coding-Ready (Explicit Contracts & Complete Seed)	Timer Governance:	20s System Default (Director Auth Required)

1. Document Purpose, System Overview & Technology Stack Matrix

This Detailed Design Document (DDD v1.3) provides the exact implementation contracts for autonomous AI coding agents (Gemini and Claude). It eliminates all architectural ambiguities by detailing binary structures, WebRTC signaling schemas, full database DDL with seed data, and a mock stylus simulator.

2. Component Architecture, Data Flow, and Dual WebRTC Channels

- **Dual WebRTC Channels:** Channel 1 (emr-stream): Unordered/unreliable SCTP for 133Hz coordinate streaming (<15ms latency). Channel 2 (control-events): Ordered/reliable SCTP for lifecycle commands, timers, and alerts.

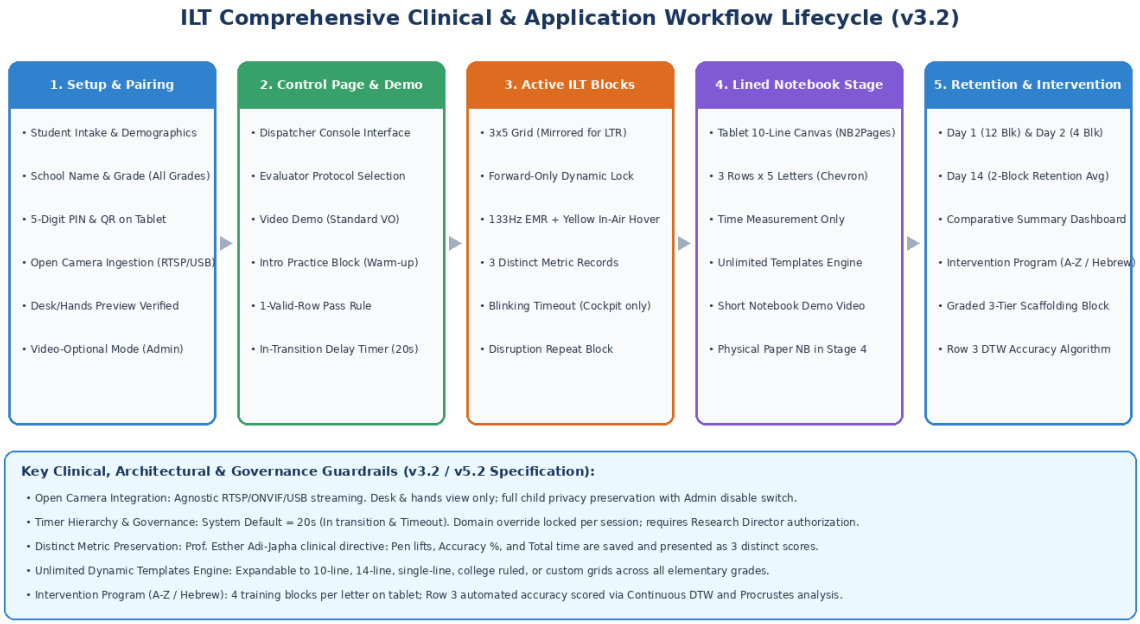


Figure 1: Comprehensive Clinical and System Lifecycle Workflow

3. Monorepo Project Structure & Module Organization

```
ilt-research-platform/
├── apps/
│   ├── student-tablet/          # Tablet Kiosk App (React + Canvas 2D + EMR Driver)
│   ├── proctor-cockpit/         # Evaluator Console (Control Dispatcher, Live Cockpit, Comparative
Dashboard)
│   └── analyst-portal/          # Analyst Portal (Kinematic Replayer, Desk Video Sync, Superset
Dashboards)
├── services/
│   ├── gateway/                 # Local Gateway (Signaling, DB Cache, go2rtc Camera Proxy)
│   └── analytics/               # Kinematic Analytics & DTW Scoring Service (Python FastAPI)
├── packages/
│   ├── shared-types/            # Shared TypeScript interfaces, data models, and enums
│   ├── emr-protocol/            # 24-byte binary serialization and OpCodes parser
│   └── template-schemas/        # JSON Schemas for dynamic rulings (10-line, 14-line, 1-line,
grids)
└── docker/                      # Development and deployment containers (TimescaleDB, MinIO,
go2rtc)
```

4. Binary Wire Protocol, Serialization & WebRTC Signaling Contracts

```
// C / C++ & TypedArray 24-Byte Binary EMR Sample Struct (Little-Endian IEEE 754)
#pragma pack(push, 1)
struct EMRSampleRecord {
    uint32_t timestamp_offset_ms; // 4 Bytes: Milliseconds elapsed since block launch
    float    x_coord;             // 4 Bytes: Normalized X sub-pixel coordinate (0.0000 - 1.0000)
    float    y_coord;             // 4 Bytes: Normalized Y sub-pixel coordinate (0.0000 - 1.0000)
    float    pressure;            // 4 Bytes: Normalized pen tip pressure (0.0000 - 1.0000)
    int16_t  tilt_altitude;       // 2 Bytes: Pen vertical angle relative to tablet plane (0 - 90
deg)
    int16_t  tilt_azimuth;        // 2 Bytes: Pen rotational angle around surface normal (0 - 360
```

```
deg)
    uint8_t  stroke_flags;        // 1 Byte: Bit 0: PenDown, Bit 1: InAirHover, Bit 2: Eraser, Bit
3: TargetHit
    uint8_t  active_house_id;     // 1 Byte: Identifier of active house/cell (1 to 15)
    uint16_t reserved_padding;    // 2 Bytes: Alignment padding (0x0000)
};
#pragma pack(pop)
```

WebRTC Signaling Protocol Contracts (WebSocket JSON):

```
// WebRTC Signaling Protocol Contracts (WebSocket / JSON)
// 1. Tablet -> Gateway (Signaling): Join session room
{
  "event": "JOIN_ROOM",
  "payload": {
    "role": "TABLET",
    "pin": "84920",
    "device_model": "Wacom MovinkPad 11 / DTHA116"
  }
}

// 2. Gateway -> Proctor Cockpit: Notify tablet joined
{
  "event": "PEER_JOINED",
  "payload": {
    "peer_id": "tablet_client_uuid",
    "role": "TABLET"
  }
}

// 3. Proctor Cockpit -> Gateway -> Tablet: WebRTC SDP Offer
{
  "event": "WEBRTC_OFFER",
  "payload": {
    "sdp": "v=0\r\no=- 420 2 IN IP4 127.0.0.1...",
    "type": "offer"
  }
}

// 4. Tablet -> Gateway -> Cockpit: WebRTC SDP Answer
{
  "event": "WEBRTC_ANSWER",
  "payload": {
    "sdp": "v=0\r\no=- 421 2 IN IP4 127.0.0.1...",
    "type": "answer"
  }
}

// 5. Bidirectional: ICE Candidate Exchange
{
  "event": "ICE_CANDIDATE",
  "payload": {
    "candidate": "candidate:842163049 1 udp 1677729535 192.168.1.105 52834 typ host...",
    "sdpMid": "0",
    "sdpMLineIndex": 0
```

```
  }
}
```

5. Dynamic Canvas Engine, Template Geometry, and Forward-Only Lock

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "ILTDynamicTemplateDefinition",
  "type": "object",
  "properties": {
    "template_id": { "type": "string" },
    "template_name": { "type": "string" },
    "ruling_type": { "type": "string", "enum": ["NOTEBOOK_10_ROWS", "NOTEBOOK_14_ROWS",
"NOTEBOOK_1_ROW", "GRID_3X5", "FREE_CANVAS"] },
    "target_grade_level": { "type": "string", "default": "ALL_GRADES" },
    "canvas_schema": {
      "type": "object",
      "properties": {
        "num_lines": { "type": "integer" },
        "line_spacing_px": { "type": "number" },
        "margin_side": { "type": "string", "enum": ["RIGHT", "LEFT", "NONE"] },
        "margin_offset_px": { "type": "number" },
        "active_row_indices": { "type": "array", "items": { "type": "integer" } }
      },
      "required": ["num_lines", "line_spacing_px", "margin_side", "active_row_indices"]
    },
    "rules_schema": {
      "type": "object",
      "properties": {
        "max_internal_pen_lifts": { "type": "integer", "default": 0 },
        "width_nominal_mm": { "type": "number", "default": 15.0 },
        "width_tolerance_mm": { "type": "number", "default": 2.0 },
        "forward_only_locking": { "type": "boolean", "default": true }
      }
    }
  },
  "required": ["template_id", "template_name", "ruling_type", "canvas_schema", "rules_schema"]
}
```

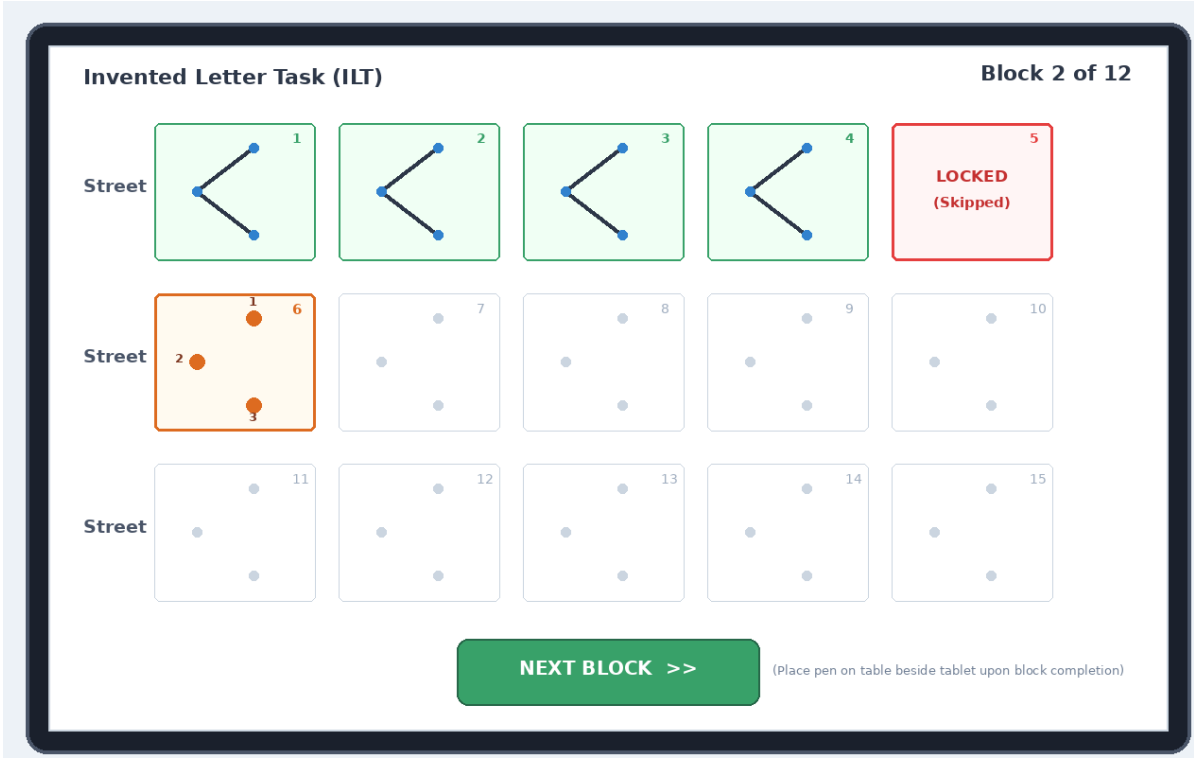


Figure 2: Student Tablet Canvas in LTR Mode – House 5 Locked (Skipped), House 6 Active

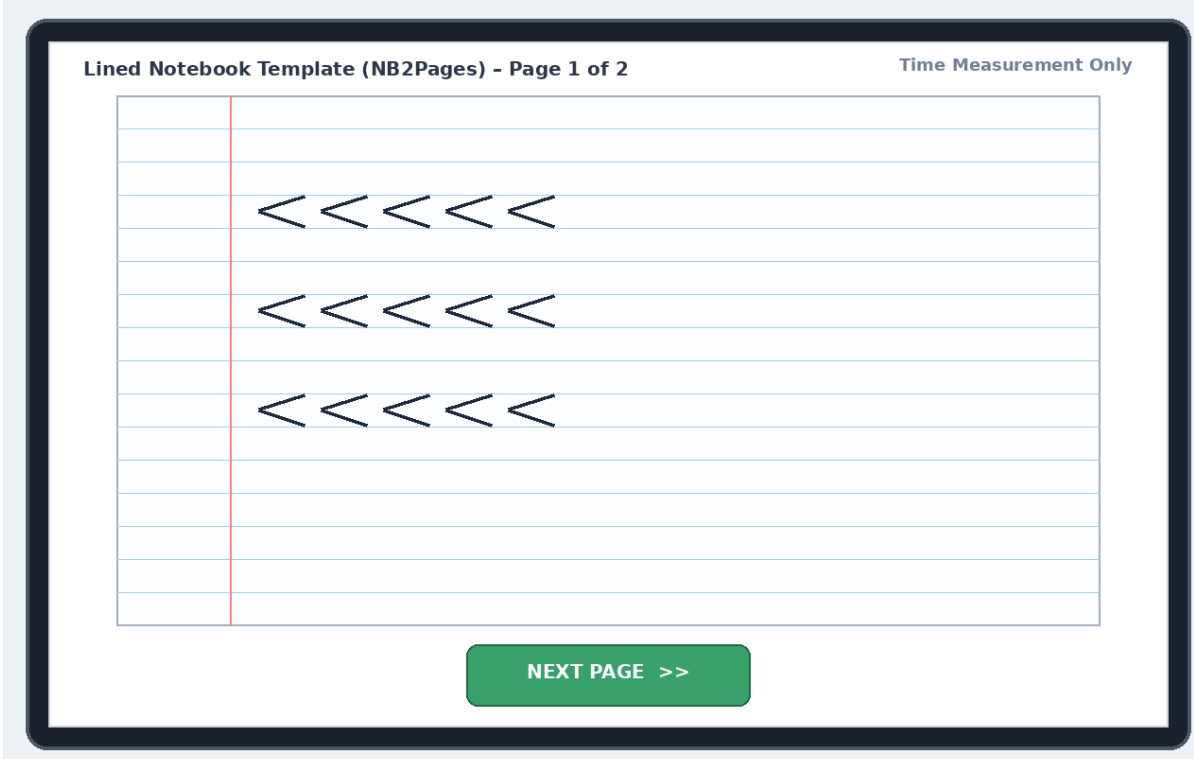


Figure 3: Lined Notebook Canvas Aligned to Original Study Ruling (3 Rows of 5 Chevrons)

6. Finite State Machines & Protocol Workflow Execution

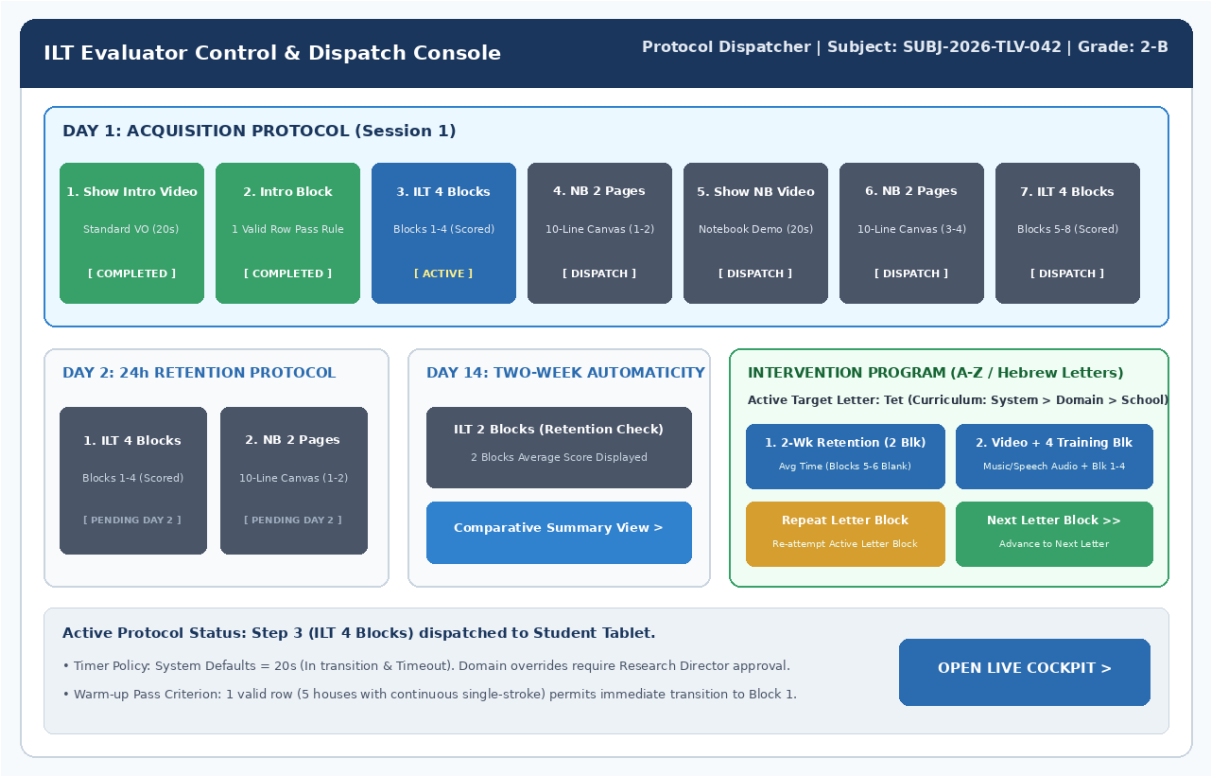


Figure 4: Evaluator Control & Protocol Dispatcher Console with Intervention Program Rubric



Figure 5: Proctor Cockpit (80% Live Mirror / 20% Clinical Telemetry) with Access to Comparative Summary

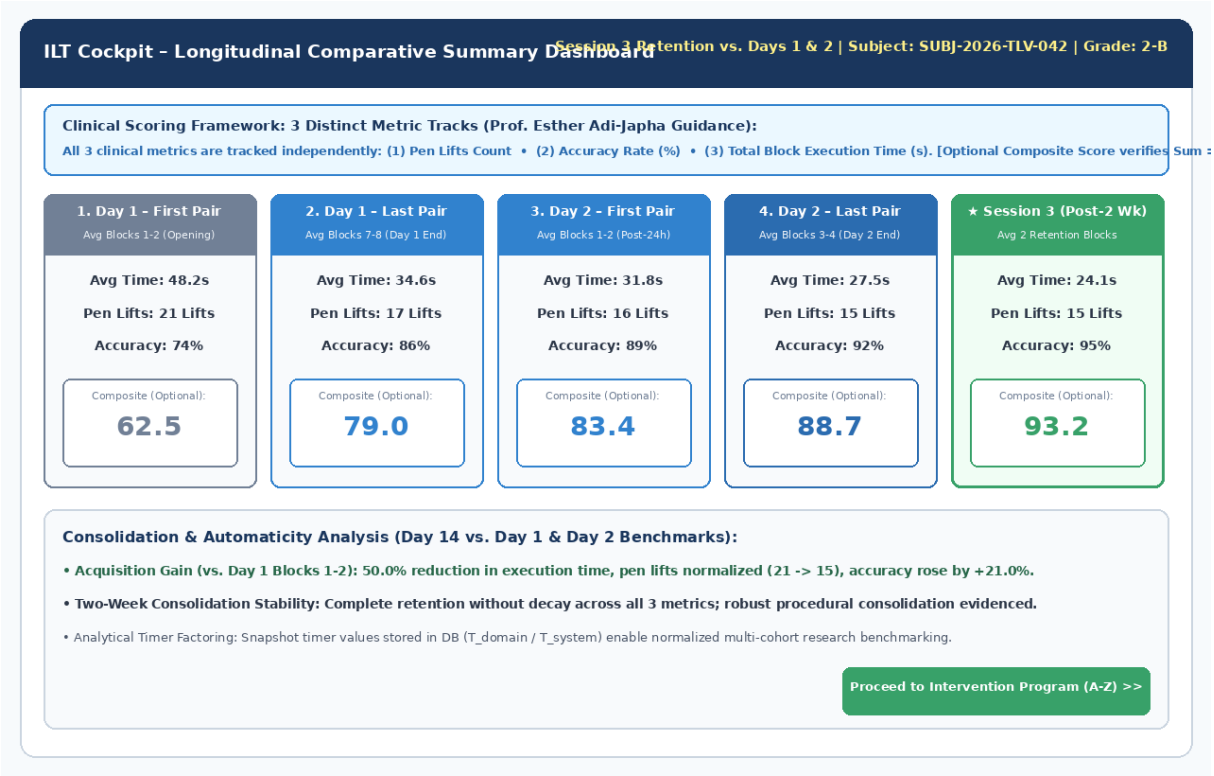


Figure 6: Longitudinal Comparative Summary Dashboard Displaying 3 Distinct Clinical Metrics

7. Physical Database Schema & Seed Data (PostgreSQL 16 + TimescaleDB)

```
-- ================================================================================
-- ILT PRODUCTION DDL (PostgreSQL 16 + TimescaleDB) - Release 5.3 / DDD v1.2
-- ================================================================================
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS "timescaledb";

-- 1. System Configuration & Global Defaults
CREATE TABLE system_config (
    config_id INT PRIMARY KEY DEFAULT 1,
    in_transition_delay_sec INT NOT NULL DEFAULT 20, -- Default In-Transition: 20s
    inactivity_timeout_sec INT NOT NULL DEFAULT 20,  -- Default Inactivity Timeout: 20s
    post_block_timeout_sec INT NOT NULL DEFAULT 30,  -- Timeout post-15th letter: 30s
    notebook_page_timeout_sec INT NOT NULL DEFAULT 60,
    video_recording_enabled BOOLEAN NOT NULL DEFAULT TRUE,
    require_research_director_approval BOOLEAN NOT NULL DEFAULT TRUE,
    score_weight_pen_lifts_pct NUMERIC(5,2) NOT NULL DEFAULT 35.00,
    score_weight_accuracy_pct NUMERIC(5,2) NOT NULL DEFAULT 40.00,
    score_weight_time_pct NUMERIC(5,2) NOT NULL DEFAULT 25.00,
    CONSTRAINT chk_weights_sum_100 CHECK (
        (score_weight_pen_lifts_pct + score_weight_accuracy_pct + score_weight_time_pct) = 100.00
    ),
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 2. Domains Table (with Research Director Authorization for Overrides)
CREATE TABLE domains (
    domain_id SERIAL PRIMARY KEY,
```

```
    domain_name VARCHAR(100) NOT NULL UNIQUE,
    override_in_transition_sec INT NULL,
    override_inactivity_timeout_sec INT NULL,
    override_video_enabled BOOLEAN NULL,
    timer_approved_by_research_director VARCHAR(100) NULL,
    timer_approval_timestamp TIMESTAMP WITH TIME ZONE NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 3. Dynamic Templates Engine
CREATE TABLE dynamic_templates (
    template_id VARCHAR(50) PRIMARY KEY,
    template_name VARCHAR(100) NOT NULL,
    template_category VARCHAR(30) NOT NULL,
    ruling_type VARCHAR(30) NOT NULL,
    target_grade_level VARCHAR(30) DEFAULT 'ALL_GRADES',
    canvas_schema JSONB NOT NULL,
    rules_schema JSONB NOT NULL,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 4. Canonical Models for Freehand Row 3 Evaluation
CREATE TABLE letter_canonical_models (
    letter_id VARCHAR(10) PRIMARY KEY,
    language_code VARCHAR(5) NOT NULL,
    letter_char VARCHAR(5) NOT NULL,
    expected_strokes_count INT NOT NULL DEFAULT 1,
    canonical_aspect_ratio NUMERIC(5,3) NOT NULL,
    canonical_points JSONB NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 5. Sessions Table (3 Metrics Preserved Independently + Timer Snapshots)
CREATE TABLE sessions (
    session_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_number BIGSERIAL UNIQUE,
    domain_id INT NOT NULL REFERENCES domains(domain_id),
    school_name VARCHAR(150) NOT NULL,
    grade_class VARCHAR(50) NOT NULL,
    evaluator_name VARCHAR(120) NOT NULL,
    assessment_date DATE NOT NULL DEFAULT CURRENT_DATE,
    examinee_alias VARCHAR(50) NOT NULL,
    examinee_initials VARCHAR(5) NOT NULL,
    student_language VARCHAR(5) DEFAULT 'he',
    session_stage VARCHAR(30) NOT NULL,
    is_charging_during_test BOOLEAN DEFAULT FALSE,
    sys_in_transition_sec INT NOT NULL DEFAULT 20,
    dom_in_transition_sec INT NOT NULL DEFAULT 20,
    sys_inactivity_sec INT NOT NULL DEFAULT 20,
    dom_inactivity_sec INT NOT NULL DEFAULT 20,
    video_recording_enabled BOOLEAN NOT NULL DEFAULT TRUE,
    warmup_row_passed BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
```

```
);

-- 6. Assessment Blocks Table (Preserving 3 Distinct Metrics Independently)
CREATE TABLE session_blocks (
    block_id BIGSERIAL PRIMARY KEY,
    session_id UUID NOT NULL REFERENCES sessions(session_id) ON DELETE CASCADE,
    block_type VARCHAR(30) NOT NULL,
    block_index INT NOT NULL,
    target_letter VARCHAR(5) NULL,
    pen_lifts_count INT NOT NULL,
    accuracy_pct NUMERIC(5,2) NOT NULL,
    total_duration_ms INT NOT NULL,
    composite_score NUMERIC(6,2) NULL,
    is_repeated_disrupted BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 7. High-Frequency EMR Telemetry (TimescaleDB Hypertable with 1-Day Chunks)
CREATE TABLE pen_telemetry_chunks (
    recorded_at TIMESTAMP WITH TIME ZONE NOT NULL,
    session_id UUID NOT NULL,
    block_index INT NOT NULL,
    house_id INT NOT NULL,
    time_offset_ms INT NOT NULL,
    x NUMERIC(8,4) NOT NULL,
    y NUMERIC(8,4) NOT NULL,
    pressure NUMERIC(6,5) NOT NULL,
    tilt_altitude SMALLINT NOT NULL,
    tilt_azimuth SMALLINT NOT NULL,
    is_hover BOOLEAN NOT NULL,
    is_pen_down BOOLEAN NOT NULL
);
SELECT create_hypertable('pen_telemetry_chunks', 'recorded_at', chunk_time_interval => INTERVAL '1
day');
ALTER TABLE pen_telemetry_chunks SET (timescaledb.compress, timescaledb.compress_segmentby =
'session_id, block_index');
SELECT add_compression_policy('pen_telemetry_chunks', INTERVAL '7 days');

-- 8. Video Replay Records (Desk & Hands RTSP / MP4 Ingestion)
CREATE TABLE session_video_records (
    video_id BIGSERIAL PRIMARY KEY,
    session_id UUID NOT NULL REFERENCES sessions(session_id) ON DELETE CASCADE,
    camera_model VARCHAR(100) DEFAULT 'Wireless IP Camera (RTSP)',
    storage_bucket VARCHAR(50) DEFAULT 'ilt-session-videos',
    video_object_key TEXT NOT NULL,
    duration_sec NUMERIC(8,2) NOT NULL,
    start_time_utc TIMESTAMP WITH TIME ZONE NOT NULL,
    fps NUMERIC(4,2) DEFAULT 25.00,
    resolution VARCHAR(20) DEFAULT '1280x720',
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 9. Analytical Cohort Slices View (for Apache Superset, Metabase, DuckDB, Jupyter)
CREATE OR REPLACE VIEW v_analyst_cohort_slices AS
```

```
SELECT
    s.session_id,
    s.session_number,
    s.examinee_alias,
    s.examinee_initials,
    d.domain_name,
    s.school_name,
    s.grade_class,
    s.dominant_hand,
    s.student_language,
    s.session_stage,
    s.assessment_date,
    s.sys_in_transition_sec,
    s.dom_in_transition_sec,
    (s.dom_in_transition_sec::numeric / s.sys_in_transition_sec::numeric) AS timer_factoring_ratio,
    sb.block_id,
    sb.block_type,
    sb.block_index,
    sb.target_letter,
    sb.pen_lifts_count,
    sb.accuracy_pct,
    sb.total_duration_ms,
    (sb.total_duration_ms::numeric / 1000.0) AS total_duration_sec,
    ((sb.total_duration_ms::numeric / 1000.0) * (s.sys_in_transition_sec::numeric /
s.dom_in_transition_sec::numeric)) AS factored_duration_sec,
    sb.composite_score,
    sb.is_repeated_disrupted
FROM sessions s
JOIN domains d ON s.domain_id = d.domain_id
JOIN session_blocks sb ON s.session_id = sb.session_id;
```

Seed Data Script (Timers: 20s Default, Preconfigured Domains & Templates):

```
-- ============================================================================
-- SEED DATA & INITIAL CONFIGURATION (PostgreSQL 16)
-- ============================================================================

-- 1. Insert System Default Configuration (Timers: 20s, Sum(weights) = 100%)
INSERT INTO system_config (
    config_id, in_transition_delay_sec, inactivity_timeout_sec, post_block_timeout_sec,
    notebook_page_timeout_sec, video_recording_enabled, require_research_director_approval,
    score_weight_pen_lifts_pct, score_weight_accuracy_pct, score_weight_time_pct
) VALUES (
    1, 20, 20, 30, 60, TRUE, TRUE, 35.00, 40.00, 25.00
) ON CONFLICT (config_id) DO NOTHING;

-- 2. Insert Default Districts (Domains)
INSERT INTO domains (domain_name, override_in_transition_sec, override_inactivity_timeout_sec)
VALUES
    ('Central District (מחוז מרכז)', NULL, NULL),
    ('Tel Aviv District (מחוז תל אביב)', NULL, NULL),
    ('Haifa District (מחוז חיפה)', NULL, NULL),
    ('Jerusalem District (מחוז ירושלים)', NULL, NULL),
    ('Northern District (מחוז צפון)', NULL, NULL),
    ('Southern District (מחוז דרום)', NULL, NULL)
```

```
ON CONFLICT (domain_name) DO NOTHING;

-- 3. Insert Standard Dynamic Templates
INSERT INTO dynamic_templates (
    template_id, template_name, template_category, ruling_type, target_grade_level,
    canvas_schema, rules_schema
) VALUES
(
    'TEMPLATE_ILT_GRID_3X5',
    'Standard ILT 3x5 Grid (רשת 15 בתים לאות מומצאת)',
    'GRID_ILT',
    'GRID_3X5',
    'ALL_GRADES',
    '{"rows": 3, "cols": 5, "box_w_px": 170, "box_h_px": 145, "dot_hit_radius_mm": 3.5}',
    '{"max_internal_pen_lifts": 0, "width_nominal_mm": 15.0, "width_tolerance_mm": 2.0,
"forward_only_locking": true}'
),
(
    'TEMPLATE_NOTEBOOK_10_ROWS',
    'Authentic 10-Row Israeli Elementary Notebook (מחברת 10 שורות לכיתות יסודי)',
    'NOTEBOOK_LINED',
    'NOTEBOOK_10_ROWS',
    'ALL_GRADES',
    '{"num_lines": 16, "line_spacing_px": 35.0, "margin_side": "RIGHT", "active_row_indices": [4,
7, 10]}',
    '{"max_internal_pen_lifts": 0, "time_measurement_only": true, "letters_per_row": 5}'
),
(
    'TEMPLATE_INTERVENTION_LETTER_TET',
    'Intervention Scaffolding Template (האות העברית ט'')',
    'INTERVENTION_LETTER',
    'GRID_3X5',
    'ALL_GRADES',
    '{"row1": "MODEL_PLUS_TARGET_DOTS", "row2": "START_DOT_ONLY", "row3": "BLANK_FREEHAND"}',
    '{"max_internal_pen_lifts": 0, "dtw_accuracy_evaluation_row3": true, "nominal_aspect_ratio":
0.85}'
)
ON CONFLICT (template_id) DO NOTHING;
```

8. Kinematic Session Replay Engine & Cohort Analytics Implementation

```
// Kinematic Session Replay Engine (TypeScript / React Canvas 2D)
export class KinematicSessionReplayer {
    private currentFrameIndex = 0;
    private playbackSpeed = 1.0;
    private isPlaying = false;
    private animFrameId: number | null = null;

    constructor(
        private canvas: HTMLCanvasElement,
        private samples: any[],
        private videoElement?: HTMLVideoElement,
        private onProgress?: (timeMs: number, sample: any) => void
    ) {}
```

```
    public play(): void {
        this.isPlaying = true;
        if (this.videoElement) this.videoElement.play();
        this.renderLoop();
    }

    public pause(): void {
        this.isPlaying = false;
        if (this.animFrameId) cancelAnimationFrame(this.animFrameId);
        if (this.videoElement) this.videoElement.pause();
    }

    public setSpeed(speed: number): void {
        this.playbackSpeed = speed;
        if (this.videoElement) this.videoElement.playbackRate = speed;
    }

    public seek(targetTimeMs: number): void {
        this.currentFrameIndex = this.samples.findIndex(s => s.timestamp_offset_ms >=
targetTimeMs);
        if (this.currentFrameIndex === -1) this.currentFrameIndex = this.samples.length - 1;
        if (this.videoElement) this.videoElement.currentTime = targetTimeMs / 1000.0;
        this.renderUpToCurrentFrame();
    }

    private renderLoop = (): void => {
        if (!this.isPlaying) return;
        const currentSample = this.samples[this.currentFrameIndex];
        this.drawStrokeSegment(currentSample);
        if (this.onProgress) this.onProgress(currentSample.timestamp_offset_ms, currentSample);

        this.currentFrameIndex += Math.max(1, Math.round(this.playbackSpeed));
        if (this.currentFrameIndex < this.samples.length) {
            this.animFrameId = requestAnimationFrame(this.renderLoop);
        } else {
            this.pause();
        }
    };

    private drawStrokeSegment(s: any): void {
        const ctx = this.canvas.getContext('2d')!;
        const x = s.x_coord * this.canvas.width;
        const y = s.y_coord * this.canvas.height;
        if (s.stroke_flags & 0x01) { // Pen Down
            ctx.fillStyle = this.getPressureColor(s.pressure);
            ctx.beginPath();
            ctx.arc(x, y, 2 + s.pressure * 4, 0, Math.PI * 2);
            ctx.fill();
        } else if (s.stroke_flags & 0x02) { // In-Air Hover
            ctx.fillStyle = '#ECC94B'; // Yellow hover line
            ctx.fillRect(x - 1, y - 1, 2, 2);
        }
    }
```

```
    private getPressureColor(p: number): string {
        if (p < 0.20) return '#38A169'; // Light Green
        if (p < 0.40) return '#3182CE'; // Medium Blue
        return '#E53E3E';               // Firm Red
    }
}
```

```
# Python Cohort Analytics & Parquet / DuckDB Export Engine (FastAPI)
import duckdb
import pandas as pd
from fastapi import APIRouter, Response

router = APIRouter(prefix="/api/v1/analyst")

@router.get("/export/parquet")
def export_cohort_parquet(domain: str = None, grade: str = None, ses_low: bool = None):
    con = duckdb.connect()
    con.execute("INSTALL postgres; LOAD postgres;")
    con.execute("ATTACH 'dbname=ilt_research user=ilt_admin host=db' AS pg (TYPE POSTGRES);")

    query = "SELECT * FROM pg.v_analyst_cohort_slices WHERE 1=1"
    if domain: query += f" AND domain_name = '{domain}'"
    if grade: query += f" AND grade_class LIKE '%{grade}%'"

    df = con.execute(query).df()
    parquet_bytes = df.to_parquet(index=False)

    return Response(
        content=parquet_bytes,
        media_type="application/octet-stream",
        headers={"Content-Disposition": "attachment; filename=ilt_cohort_data.parquet"}
    )
```

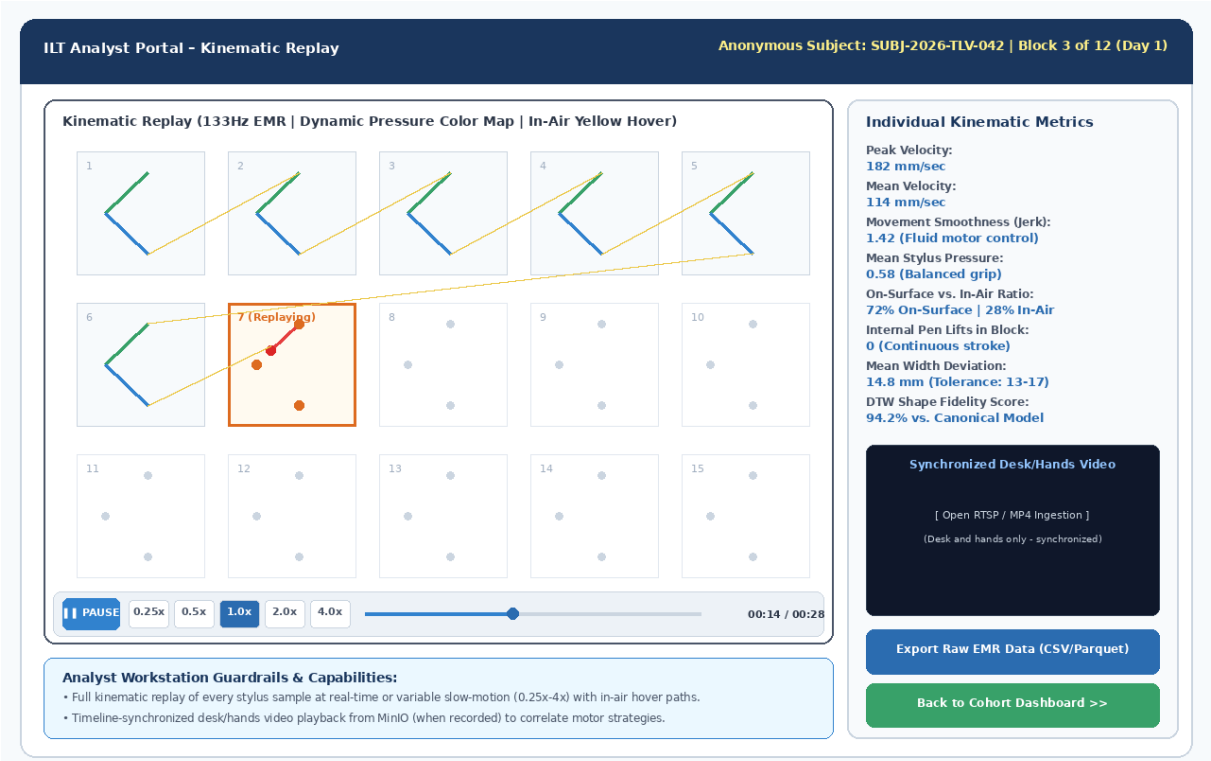


Figure 7: Analyst Portal – Kinematic Session Replay Engine Synced with Desk Video

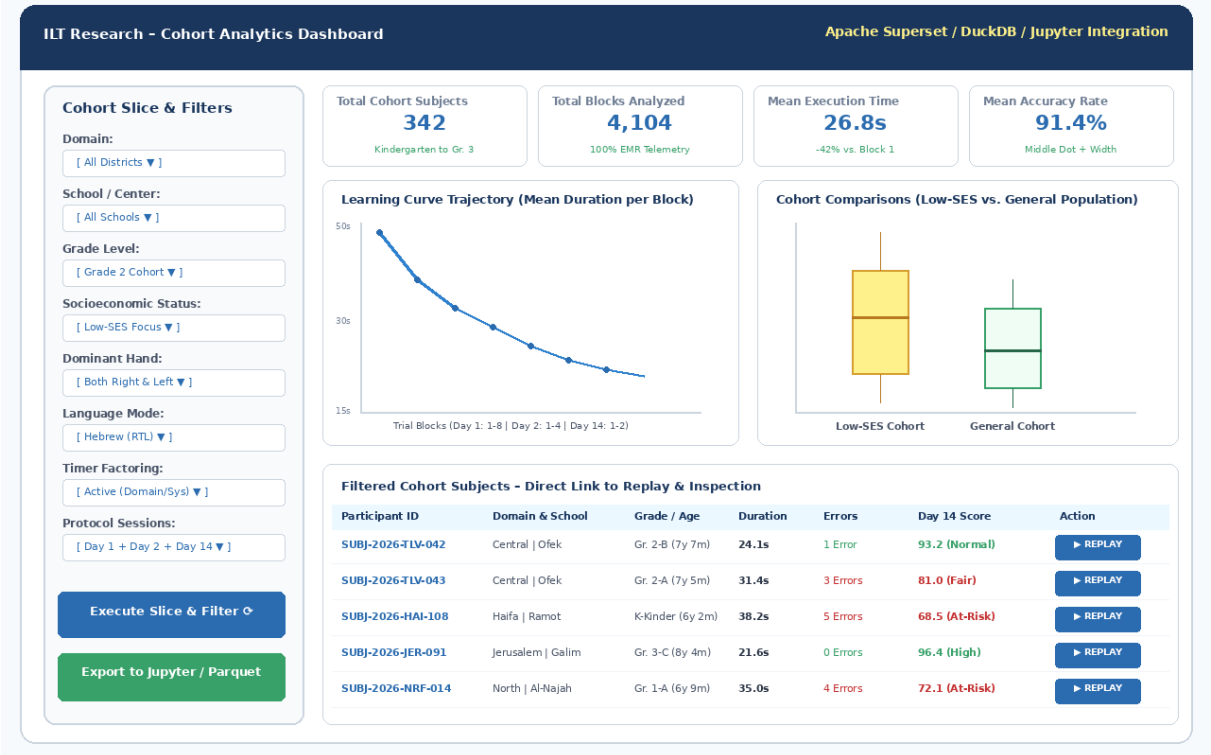


Figure 8: Open-Source Cohort Slice-and-Dice Analytics Dashboard

9. Kinematic Algorithms & Intervention Program

```
import numpy as np
from scipy.spatial import procrustes
```

```
from fastdtw import fastdtw

def evaluate_freehand_row3(stroke_points: np.ndarray, canonical_points: np.ndarray,
canonical_aspect_ratio: float) -> dict:
    if len(stroke_points) < 10:
        return {"composite_accuracy": 0.0, "diagnostic_tag": "EMPTY_OR_NOISE", "valid": False}

    diffs = np.diff(stroke_points, axis=0)
    dists = np.hypot(diffs[:, 0], diffs[:, 1])
    cum_dist = np.insert(np.cumsum(dists), 0, 0)
    total_len = cum_dist[-1]
    even_spaces = np.linspace(0, total_len, 100)
    resampled_stroke = np.column_stack((
        np.interp(even_spaces, cum_dist, stroke_points[:, 0]),
        np.interp(even_spaces, cum_dist, stroke_points[:, 1])
    ))

    norm_stroke = (resampled_stroke - np.mean(resampled_stroke, axis=0)) /
(np.std(resampled_stroke) + 1e-6)
    norm_canonical = (canonical_points - np.mean(canonical_points, axis=0)) /
(np.std(canonical_points) + 1e-6)

    _, _, disparity = procrustes(norm_canonical, norm_stroke)
    procrustes_score = max(0.0, min(100.0, 100.0 * (1.0 - disparity * 2.2)))

    dtw_dist, _ = fastdtw(norm_stroke, norm_canonical, dist=2)
    dtw_score = max(0.0, min(100.0, 100.0 * (1.0 - (dtw_dist / 100.0))))

    w, h = np.ptp(stroke_points[:, 0]), np.ptp(stroke_points[:, 1])
    actual_ratio = w / (h + 1e-6)
    ratio_dev_pct = abs(actual_ratio - canonical_aspect_ratio) / canonical_aspect_ratio * 100.0

    composite_acc = 0.55 * procrustes_score + 0.30 * dtw_score + 0.15 * max(0.0, 100.0 -
ratio_dev_pct)
    tag = "ACCURATE" if composite_acc >= 80.0 else ("PROPORTION_DISTORTION" if ratio_dev_pct > 25.0
else "SHAPE_DEFORMATION")

    return {
        "composite_accuracy": round(composite_acc, 2),
        "procrustes_score": round(procrustes_score, 2),
        "dtw_distance": round(dtw_dist, 2),
        "aspect_ratio_deviation_pct": round(ratio_dev_pct, 2),
        "diagnostic_tag": tag,
        "valid": True
    }
```

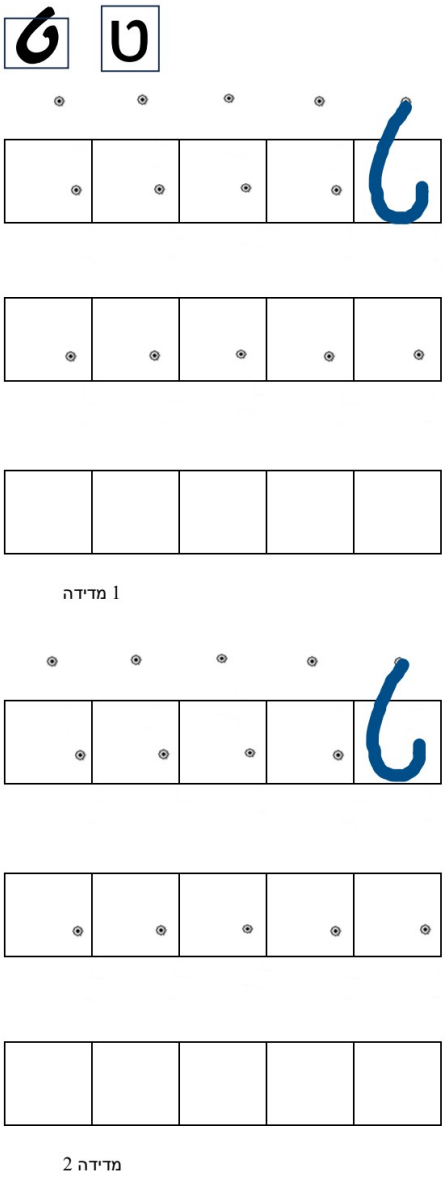


Figure 9: Sample Graded Intervention Training Sheet (Hebrew Letter Tet)

10. Appendix: Developer Implementation & AI Coding Guide

```
version: '3.8'
services:
  db:
    image: timescale/timescaledb:latest-pg16
    container_name: ilt_db
    environment:
      POSTGRES_DB: ilt_research
      POSTGRES_USER: ilt_admin
      POSTGRES_PASSWORD: ilt_secret_password
    ports:
      - "5432:5432"
    volumes:
      - pgdata:/var/lib/postgresql/data
    restart: always
```

```
  minio:
    image: minio/minio:latest
    container_name: ilt_minio
    command: server /data --console-address ":9001"
    environment:
      MINIO_ROOT_USER: minio_admin
      MINIO_ROOT_PASSWORD: minio_secret_password
    ports:
      - "9000:9000"
      - "9001:9001"
    volumes:
      - miniodata:/data
    restart: always

  go2rtc:
    image: alexxit/go2rtc:latest
    container_name: ilt_cms_gateway
    ports:
      - "1984:1984" # Management & WebRTC API
      - "8554:8554" # RTSP Relay Port
      - "8555:8555" # WebRTC Live Port
    volumes:
      - ./go2rtc.yaml:/config/go2rtc.yaml
    restart: always

volumes:
  pgdata:
  miniodata:
```

```
// Mock Wacom EMR Stylus Simulator for Desktop Browser Development
export class MockEMRStylusSimulator {
    private isMouseDown = false;
    private startTime = 0;

    constructor(private canvas: HTMLCanvasElement, private onSample: (s: any) => void) {
        this.canvas.addEventListener('mousedown', (e) => {
            this.isMouseDown = true;
            this.startTime = performance.now();
            this.emit(e, 0.45, true, false);
        });
        window.addEventListener('mouseup', (e) => {
            if (this.isMouseDown) {
                this.isMouseDown = false;
                this.emit(e, 0.0, false, false);
            }
        });
        this.canvas.addEventListener('mousemove', (e) => {
            const pressure = this.isMouseDown ? 0.55 + 0.15 * Math.sin(performance.now() / 80) :
0.0;
            const isHover = !this.isMouseDown;
            this.emit(e, pressure, this.isMouseDown, isHover);
        });
    }
```

```
    private emit(e: MouseEvent, pressure: number, isPenDown: boolean, isHover: boolean) {
        const rect = this.canvas.getBoundingClientRect();
        const x = (e.clientX - rect.left) / rect.width;
        const y = (e.clientY - rect.top) / rect.height;
        this.onSample({
            time_offset_ms: Math.round(performance.now() - this.startTime),
            x: Number(x.toFixed(5)),
            y: Number(y.toFixed(5)),
            pressure: Number(pressure.toFixed(4)),
            tilt_altitude: 60,
            tilt_azimuth: 180,
            is_pen_down: isPenDown,
            is_hover: isHover
        });
    }
}
```